

Paper 26 - Z-Pinch and Shear Horizon

CQE-paper-26

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Construction Basis

Speculatively examine first-shear, pinch, and friction-like generation as a later horizon paper requiring strict separation from proven layers.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- promoted formal paper: production\formal-papers\CQE-paper-26\FORMAL_PAPER.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-26\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-26\verify_zpinch_shear_horizon.py
- receipt: production\formal-papers\CQE-paper-26\zpinch_shear_horizon_receipt.json (CQE-paper-26: pass_with_open_obligations)
- body: production\papers\CQE-paper-26\PAPER-BODY.md
- source: production\papers\CQE-paper-26\SOURCE.md
- formal: production\papers\CQE-paper-26\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-26\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-26\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-26.md

Paper 26 - Z-Pinch and Shear Horizon

Abstract

Paper 26 is a horizon paper. Its closed mathematical content is the Oloid/octonion carrier algebra and a ledger definition of pinch/shear as an analog boundary event. It does not prove a physical Z-pinch, magnetic confinement mechanism, friction mechanism, or energy-generation process.

The formal receipt verifies that the integer Oloid carrier has period-4 rolling closure, that the octonion carrier realizes the same period through $e^2 = -1$ and $e^4 = 1$, and that a fixed-generator comparison on a Rule 30 center-column tape exposes a reproducible shear analog. The pinch analog is defined as ledger reclassification toward bounded residue or open lift. Any plasma interpretation remains an explicitly named obligation.

Closed Evidence

The receipt is generated by ``production/formal-papers/CQE-paper-26/verify_zpinch_shear_horizon.py``. It imports the live package modules ``oloid_rolling``, ``oloid_octonionic``, ``rule30_nth_bit``, and ``transport_obligations``.

The receipt status is ``pass_with_open_obligations``. The integer rolling verifier passes with ``bit0_period_4 = true``, ``bit1_period_4 = true``, invariant sheet and phase fractions equal to ``1.0`` at ``K = 6``, and a ``K = 8`` landing table of ``256`` entries. The octonion verifier passes with ``e4_squared_is_minus_one = true``, ``e4_fourth_is_one = true``, and ``non_associative_imaginary_units = true``.

The worked probe reads the first sixteen Rule 30 center-column bits:

``1,0,1,1,1,0,0,1,1,0,0,1,0,1,1``. The integer landing is ``(0,0,1)`` with head-tail dyad ``(0,1)``.

The fixed-generator octonion comparison produces seventeen trace rows including the initial state and reports eight nonzero shear rows. The orient-information probe over all 256 eight-bit sequences has ``trivial_baseline_rate = 0.5``, so the orient bit is not promoted into a physical observable.

Definitions

An Oloid carrier state is ``(sheet, phase, parity)``. ``sheet`` is the visible contact sheet, ``phase`` is the quarter-rotation phase modulo four, and ``parity`` is the cumulative input-bit parity.

An octonion carrier state carries an actual octonion. A bit-0 quarter step is right multiplication by e ; a bit-1 quarter step is right multiplication by e^5 . The identity $e^2 = -1$ is the gauge inversion and $e^4 = 1$ is the period-4 return.

The shear analog is the XOR divergence between two fixed-generator carrier orient bits on the same tape. It is a carrier comparison, not a plasma observable.

The pinch analog is a ledger reclassification. A demonstrated row is not pinched. A bounded row carries residue. A registered or open lift is held as open until a witness is supplied.

Claims

- The integer Oloid carrier closes as a finite period-4 rolling algebra.
- The octonion carrier realizes the same quarter-period structure with live octonion multiplication.
- The orient bit and dominant basis index give replayable path-history residue at the carrier layer.
- A fixed-generator comparison on the same tape gives a reproducible shear analog.
- A pinch in this paper is a transport-ledger reclassification, not a physical pinch event.
- Physical Z-pinch, shear-layer, friction, or generation readings are open hypotheses and must not be treated as discharged by the carrier proof.

Theorem 26

Paper 26 is valid as a CQE horizon kernel when it separates the closed carrier algebra from the candidate physical reading: period, octonion grounding, carrier residue, shear analog, and pinch reclassification may be claimed; a plasma or energy mechanism may only be logged as an obligation with a falsifier.

Proof

The integer verifier establishes the finite carrier. Four bit-0 rolls return the carrier to its initial state, and four bit-1 rolls do the same. The complement test preserves sheet and phase while carrying parity difference. The $K = 8$ table has exactly 256 entries, so every eight-bit input has a replayable landing.

The octonion verifier establishes the algebraic grounding. $e_4 * e_4$ is the negative real unit, e_4^4 is the positive real unit, and four bit-0 octonion rolls return to the initial state. The verifier also confirms non-associativity for imaginary units, so the carrier is not merely a scalar phase counter.

The shear probe then uses the first sixteen Rule 30 center bits as a common tape. One fixed carrier uses the e_4 generator and the other uses the e_5 generator. Their orient bits diverge on eight trace rows. This proves a replayable carrier-level shear analog. It does not prove that any plasma shear observable exists.

Finally, the transport rows are classified as demonstrated, bounded residue, or registered/open lift. The receipt marks all physical pinch claims false. Thus the paper closes the carrier and ledger theorem while preserving the physical reading as an obligation.

Worked Example

For the 16-bit Rule 30 center tape, the integer carrier lands at $(0,0,1)$. The fixed-generator comparison emits one row per state in the trace, including the initial row. At every row, the receipt records the input bit, the e_4 orient bit, the e_5 orient bit, their XOR shear bit, and the dominant basis index for each carrier.

The solved result is the trace itself. Eight rows have nonzero shear divergence. That is enough to use the carrier as a diagnostic map. It is not enough to call the trace a physical Z-pinch.

Open Obligations

A physical Z-pinch witness is open. A controlled plasma observable must be connected to the carrier shear bit before any physical claim can be promoted.

Friction and generation mechanisms are not claimed. Carrier residue is algebraic path history, not a mechanical energy source.

Pinch as physical collapse is not claimed. In this paper, pinch means reclassification toward residue or open lift.

A bounded-external measurement row is required before the horizon reading can leave candidate status.

Suite Role

Paper 26 demonstrates how the suite should handle high-speculation applied ideas. The paper may still be useful: it exports a real carrier residue test, a shear diagnostic, and a strict physical-claim quarantine. Later papers can use the diagnostic shape, but they inherit the obligation boundary.